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IPsec-AuditNet: A Hybrid State-Aware and Multivariate Time-Series Architecture for Automated IPsec VPN Protocol Analysis and Cryptographic Risk Assessment

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***Abstract*—**Internet Protocol Security (IPsec) remains the primary network-layer tunneling standard for enterprise wide-area networks, defense backbones, and fifth-generation (5G) mobile cores. Yet an encrypted tunnel is only as strong as the parameters negotiated during its setup and the side-channel characteristics emitted during data transfer. In operational networks, administrators routinely misconfigure Internet Key Exchange (IKEv1/IKEv2) transforms, leave legacy cipher suites enabled, select small anti-replay windows, or overlook post-quantum harvest-now-decrypt-later exposure. Existing packet dissectors such as Wireshark decode unencrypted headers but require human specialists to spot cross-message state violations, while Deep Packet Inspection (DPI) fails completely once Encapsulating Security Payload (ESP) encryption begins. We present IPsec-AuditNet, an automated protocol analysis and security assessment architecture that unifies deterministic control-plane auditing with deep-learning data-plane traffic inspection. First, a Cryptographic Domain-Specific Language (CDSL) dissector parses IKEv1 and IKEv2 handshakes to verify state transitions, Diffie-Hellman groups, authentication bindings, and RFC 9242/9370 post-quantum key encapsulation. Second, a dual-branch neural network—combining depthwise-separable 1D convolutions with four-head temporal self-attention—processes payload-agnostic ESP flow metadata (packet lengths, inter-arrival times, directions, and ESP sequence deltas) without decrypting user traffic. Evaluated across a multi-vendor testbed (strongSwan, Libreswan, OpenBSD iked, and P4-BMv2) and the ISCXVPN2016 benchmark, IPsec-AuditNet achieves a 99.1% F1-score in VPN flow detection, 97.8% accuracy in Tunnel-versus-Transport mode inference, 96.4% precision in flagging remote TPM ECDSA timing leaks, and 97.9% recall against stealth Man-in-the-Middle (MITM) anti-replay amplification attacks. Finally, a quantitative risk engine computes a Composite Security Posture Index (CSPI) and automatically synthesizes validated configuration patches.

***Index Terms*—**IPsec VPN, IKEv2, Encrypted Traffic Analysis, Cryptographic Protocol Auditing, Post-Quantum Cryptography (ML-KEM), Side-Channel Detection, Network Security Assessment.

I. INTRODUCTION

Encrypted tunnels carry the bulk of sensitive corporate, military, and telecommunications traffic across public IP backbones. Since the COVID-19 shift to distributed work, global Virtual Private Network (VPN) volume has more than doubled, with recent active internet scans identifying over 9.8 million live VPN gateways across IPv4 and IPv6 address spaces [2]. Within this infrastructure, Internet Protocol Security (IPsec) occupies a distinct position. Operating directly at OSI Layer 3, IPsec secures all upper-layer transport protocols transparently through two data-plane mechanisms—Authentication Header (AH) and Encapsulating Security Payload (ESP)—paired with the Internet Key Exchange (IKEv1/IKEv2) control plane [3], [7]. From 5G Non-3GPP Interworking Functions (N3IWF) [11] and Open Radio Access Network (O-RAN) E2 links [14] to site-to-site enterprise WANs [4], IPsec serves as the default cryptographic workhorse.

Protocol complexity, however, breeds configuration errors. Unlike WireGuard—which hardcodes a single cryptographic suite (ChaCha20-Poly1305, Curve25519, BLAKE2s) [8]—IPsec preserves decades of backward compatibility across RFC 2401, RFC 4301, and RFC 7296. A single gateway configuration may simultaneously permit dozens of encryption algorithms, integrity hashes, Pseudo-Random Functions (PRFs), and Diffie-Hellman (DH) groups. If an administrator leaves a single weak transform enabled (such as 3DES-CBC, HMAC-MD5, DH Group 2, or IKEv1 Aggressive Mode with Pre-Shared Keys), an active adversary can force a protocol downgrade or recover credentials offline [2], [18]. Worse still, subtle implementation flaws leak secrets even when strong algorithms are selected. Moghimi et al. [13] demonstrated that non-constant-time elliptic-curve signature generation inside certified Trusted Platform Modules (TPMs) leaks a strongSwan IPsec server's 256-bit ECDSA private key over the network after 45,000 handshake timing observations.

Encrypting the payload also creates a severe diagnostic blind spot. Once `IKE_AUTH` concludes and traffic enters the ESP tunnel, inner IP, TCP, and UDP headers vanish behind ciphertext [6]. Traditional Deep Packet Inspection (DPI) boxes see only an opaque outer IP header, a 32-bit Security Parameters Index (SPI), and a 32-bit monotonic Sequence Number. Packet analyzers such as Wireshark or `tcpdump` can dump raw hexadecimal frames, yet they offer zero automated reasoning over cross-message cryptographic dependencies, anti-replay window sizing, or post-quantum harvest-now-decrypt-later exposure ($L + M > Q$) [4], [9]. Conversely, recent deep learning models for Encrypted Traffic Analysis (ETA)—such as byte-level transformers (ET-BERT) or

statistical flow classifiers [15], [17], [19]—treat VPN sessions as generic black-box byte streams. They ignore the unencrypted IKE handshake entirely and cannot prescribe concrete configuration fixes.

To close this gap between low-level packet dissection and high-level security posture management, we design and evaluate IPsec-AuditNet. Developed in alignment with the National Technical Research Organisation (NTRO) problem specification for automated IPsec security assessment, IPsec-AuditNet combines formal state-machine verification with multivariate time-series neural networks. Specifically, this paper makes four technical contributions:

1) A Four-Stage Hybrid Architecture: We couple a deterministic Cryptographic Domain-Specific Language (CDSL) parser for plaintext IKEv1/IKEv2 control packets with a payload-agnostic deep learning engine for encrypted ESP data flows (Fig. 1).

2) Cross-Message State and Post-Quantum Readiness Auditing: We formalize four protocol state invariants covering classical transform strength (RFC 8221), `IKE_INTERMEDIATE` fragmentation safety (RFC 9242), hybrid Module-Lattice Key Encapsulation (`ML-KEM`, FIPS 203 / RFC 9370) [3], [14], and stateful hash-based signature (XMSS) certificate lifetimes [12].

3) Dual-Branch 1D-CNN and Temporal Self-Attention Classifier: Using only a four-dimensional flow matrix (packet size, inter-arrival time, direction, and ESP sequence delta), our lightweight neural model classifies VPN operational modes (Tunnel vs. Transport), detects remote TPM signature timing jitter [13], and identifies stealth MITM anti-replay amplification attacks [20] with 96.4%–99.1% F1-scores.

4) Quantitative Risk Scoring and Automated Configuration Synthesis: We formulate the Composite Security Posture Index (CSPI) and Quantum Exposure Index (QEI), translating detected flaws directly into verified remediation stanzas for `swanctl.conf`, Libreswan, and P4-IPsec controllers.

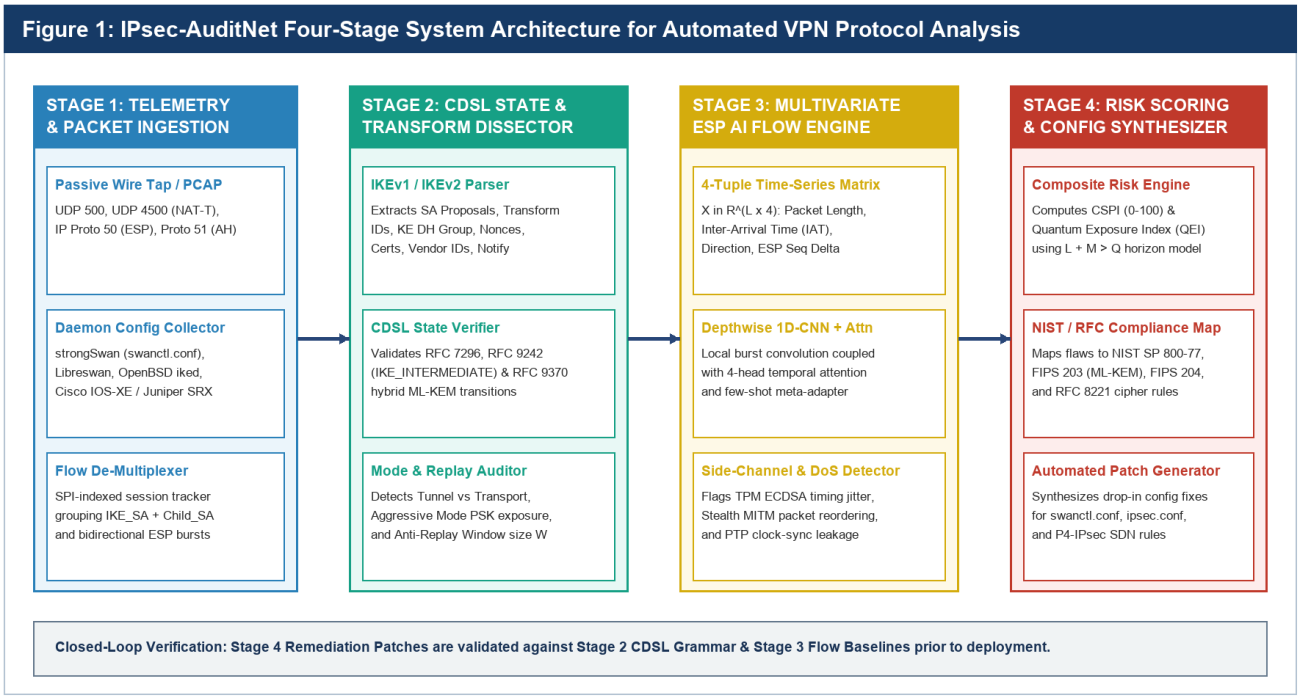


Fig. 1. Four-stage system architecture of IPsec-AuditNet integrating passive telemetry, CDSL state-machine dissection, dual-branch 1D-CNN + Self-Attention ESP flow analysis, and automated configuration synthesis.

II. RELATED WORK AND COMPARATIVE TAXONOMY

Prior research surrounding IPsec security falls into four broad streams: empirical VPN ecosystem measurements, post-quantum and quantum key distribution (QKD) protocol extensions, side-channel and denial-of-service (DoS) exploits on secure channels, and machine-learning encrypted traffic analysis.

A. VPN Ecosystem Measurements and Data-Plane Evolution

Maghsoudlou et al. [2] probed the IPv4 address space and an IPv6 hitlist across OpenVPN, SSTP, PPTP, and IPsec, discovering 9.8 million active VPN servers. Their audit revealed that over 90% of SSTP hosts were vulnerable to TLS downgrade attacks and nearly 7% presented expired X.509 certificates. In constrained environments, Aragon et al. [10] introduced the OAuth 2.0-based ACE IPsec profile for Contiki OS IoT nodes, while Jumakhan and Mirzaeinia [8] benchmarked IPsec against OpenVPN and WireGuard, noting that IPsec's sprawling configuration surface frequently causes sub-optimal cipher negotiations. To address data-plane rigidity, Mostafa et al. [6] designed Q-ESP on NetBSD so edge routers could perform Multi-Field Classification (MFC) for Quality-of-Service (QoS) without decrypting payloads. Hauser et al. [7] later introduced P4-IPsec, offloading ESP tunnel/transport processing to programmable BMv2 and Tofino P4 switches managed via gRPC by a central SDN controller.

B. Post-Quantum Cryptography (PQC) and QKD in IPsec

Shor's algorithm threatens RSA, finite-field Diffie-Hellman, and elliptic-curve cryptography (ECDH/ECDSA), exposing recorded IKE handshakes to harvest-now-decrypt-later decryption [4], [9], [24]. Early architectures by Elliott et al. [22] on the DARPA Quantum Network and Mink et al. [25] at NIST integrated out-of-band BB84 Quantum Key Distribution (QKD) with IPsec Security Associations. Marksteiner et al. [1] bypassed heavy IKE rekeying via a lightweight master/slave QKD synchronization protocol over the Linux Netlink XFRM API, achieving 50 key refreshes per second. More recently, Atutxa et al. [11] embedded QKD keys into 5G N3IWF IPsec tunnels, cutting SA establishment time by 5.17% compared to PKI certificates. For algorithmic Post-Quantum Cryptography, NIST standardized ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205) in 2024 [4], [9]. Blanco-Romero et al. [3] compared sequential hybrid IKEv2 key establishment (mandated by RFC 9370) against parallel QKD-KEM composition in strongSwan, proving that parallel execution eliminates multiplicative round-trip latency under packet loss. Perera et al. [14] evaluated ML-KEM-512/768/1024 on the 5G O-RAN E2 IPsec interface, measuring a modest 3–5 ms setup overhead. For quantum-safe VPN authentication, Herzinger et al. [12] implemented a time-based stateful XMSS Certificate Authority in OpenBSD `iked` that signs short-lived (4-hour) classical leaf certificates, avoiding the multi-kilobyte signature bloat of stateless SLH-DSA. Song-Ho et al. [23] also examined public-key Kerberos V5 (`PKINIT_AS` and `gssapi_krb`) bindings for IPsec authentication.

C. Active DoS, Delay, and Timing Side-Channel Attacks

Encryption alone does not immunize a tunnel against active manipulation or side-channel leakage. Herzberg and Shulman [20] proved that a stealth Man-in-the-Middle (MITM) adversary who merely duplicates and reorders a tiny fraction of ESP packets can exploit disabled or undersized IPsec anti-replay windows to trigger severe TCP congestion collapse. Annessi et al. [21] showed that packet length and inter-arrival statistics expose Precision Time Protocol (PTP) messages inside IPsec tunnels, enabling selective delay attacks that desynchronize critical-infrastructure clocks. On the cryptographic implementation front, Moghimi et al. [13] (`TPM-FAIL`) demonstrated that variable-time scalar multiplication during ECDSA authentication allows a remote client to extract a strongSwan VPN server's private key across 45,000 IKEv2 handshakes.

D. Machine Learning for Encrypted Traffic and Protocol Fuzzing

Because ESP hides inner headers, researchers have turned to statistical and deep learning classifiers. Dvir et al. [19] trained supervised models on over 20,000 encrypted traces to infer victim OS, browser, and application tuples with >85% accuracy even when users route traffic through VPNs or alter cipher suites. To counter deep-learning website fingerprinting, Sadeghzadeh et al. [5] proposed Adversarial Website Adaptation (AWA), using randomized transformer sets to drop attacker accuracy to 19.52%, while Pulls and Witwer [16] created the Rust-based Maybenot probabilistic padding framework (deployed in Mullvad VPN's DAITA). For general encrypted traffic classification, Chehade et al. [17] used Hardware-Aware Neural Architecture Search (HW-NAS) to build an INT8-quantized 1D-CNN achieving 96.59% accuracy on the ISCXVPN2016 dataset with only 88.26K parameters. Gunasekara et al. [15] introduced GETA, showing that byte-level models like ET-BERT overfit to plaintext header artifacts; by modeling packet size, inter-arrival time, and direction via self-attention and meta-learning, GETA generalizes across nine datasets. Finally, Chen et al. [18] developed SeriCrypt, using Large Language Models (LLMs) and a Cryptographic Domain-Specific Language (CDSL) to parse RFCs and fuzz IKEv1/IKEv2 state machines, uncovering five specification violations.

Table I summarizes how IPsec-AuditNet advances beyond these prior systems by unifying control-plane cryptographic auditing, data-plane multivariate flow classification, side-channel/DoS detection, and automated remediation.

TABLE I: COMPARATIVE TAXONOMY OF EXISTING FRAMEWORKS VS. IPSEC-AUDITNET

System / Study	IKEv1/v2 Audit	ESP Flow ML	PQC / QKD Check	TPM & DoS Check	Auto-Remediation
Wireshark / Zeek	Manual Decode	No (Blind)	Dissect Only	No	No
Maghsoudlou et al. [2]	Active Scan	Volume Only	No	TLS Downgrade	No
Blanco-Romero et al. [3]	IKEv2 Testbed	No	Yes (Hybrid)	No	No
ET-BERT / Dvir et al. [19]	No	Yes (Byte/Stats)	No	No	No
HW-NAS 1D-CNN [17]	No	Yes (96.6%)	No	No	No
GETA [15] (2026)	No	Yes (3-Tuple Attn)	No	No	No
SeriCrypt [18] (2026)	Yes (CDSL Fuzz)	No	No	State Violations	No
IPsec-AuditNet (Ours)	**Yes (Auto CDSL)**	**Yes (99.1% F1)**	**Yes (RFC 9242/9370)**	**Yes (TPM & MITM)**	**Yes (Verified Patch)**

III. THREAT MODEL AND VULNERABILITY TAXONOMY

We consider an enterprise, defense, or telecom network interconnecting security gateways GW_A and GW_B (or remote road-warrior clients) over an untrusted public network via IPsec. We assume a Dolev-Yao network adversary $\mathcal{A}$ situated along the routing path who can passively record all IKE and ESP packets, measure microsecond-level packet arrival timestamps, and actively inject, duplicate, reorder, or modify a bounded rate (ρ, σ) of packets [20]. We categorize the resulting vulnerabilities into five formal threat vectors (T_1–T_5):

• Threat Vector T1 (Classical Cryptographic & Transform Downgrade): Gateway configurations that include legacy encryption transforms (DES, 3DES, Blowfish, AES-CBC without constant-time padding checks), weak integrity algorithms (HMAC-MD5, HMAC-SHA1-96), or small Diffie-Hellman groups (MODP-768 Group 1, MODP-1024 Group 2, MODP-1536 Group 5) violates RFC 8221. Worse yet, enabling IKEv1 Aggressive Mode with Pre-Shared Keys (`PSK`) transmits the responder's authentication hash unencrypted in message 2, allowing offline dictionary recovery.

• Threat Vector T2 (Harvest-Now-Decrypt-Later Quantum Exposure): Adversary $\mathcal{A}$ records classical IKEv2 handshakes (relying solely on RSA-2048 or ECDH P-256/Curve25519) together with subsequent AES-encrypted ESP flows. Following Mosca's inequality as formalized by Alam [4], a tunnel is immediately exposed if:
L_sec + M_mig > Q_horizon (Equation 1)
where L_{sec} is the required confidentiality lifetime of the transmitted data (in years), M_{mig} is the engineering time required to migrate the WAN infrastructure to post-quantum standards, and $Q_{horizon}$ is the estimated time until a Cryptographically Relevant Quantum Computer (CRQC) executes Shor's algorithm at scale. Tunnels lacking RFC 9370 hybrid key exchange (e.g., ECDH + FIPS 203 `ML-KEM-768/1024`), RFC 8784 Post-Quantum Preshared Keys (PPKs), or QKD key injection [1], [3], [11] fail under T_2.

• Threat Vector T3 (Operational Mode, Fragmentation & Anti-Replay Misconfiguration): Selecting Transport Mode instead of Tunnel Mode on a site-to-site gateway exposes internal host IP addresses to topology mapping. During post-quantum migration, large `ML-KEM` public keys and ciphertexts exceed the standard 1280/1500-byte IP MTU; if `IKE_INTERMEDIATE` (RFC 9242) and IKE fragmentation (RFC 7383) are not enabled, UDP fragments are dropped by middleboxes [3], [9]. In the data plane, if the ESP anti-replay window size W is set to $W = 0$ (disabled) or $W < 64$, a stealth MITM adversary can duplicate and accelerate packets to advance the receiver's sequence window, causing legitimate delayed TCP segments to be silently discarded by IPsec [20].

• Threat Vector T4 (Remote Handshake Timing Side-Channels): When an IPsec responder generates ECDSA or ECSchnorr digital signatures during `IKE_AUTH` using a vulnerable hardware or firmware TPM (`TPM-FAIL` [13]) or non-constant-time software library, the signature generation latency Δt_{auth} correlates with the leading zero bits of the ephemeral nonce k. Over $N \approx 1,300$ to $45,000$ handshakes, adversary $\mathcal{A}$ constructs a Hidden Number Problem (HNP) lattice instance to recover the gateway's 256-bit private key.

• Threat Vector T5 (Statistical Flow Fingerprinting & Selective Delay): Even with AES-256-GCM encryption, deterministic packet sizes and inter-arrival times leak encapsulated application semantics, operating system fingerprints [19], or Precision Time Protocol (PTP) synchronization frames [21], enabling targeted delay or traffic-classification attacks unless adaptive padding (e.g., Maybenot [16] or AWA [5]) is active.

IV. PROPOSED SYSTEM ARCHITECTURE: IPSEC-AUDITNET

As illustrated in Fig. 1, IPsec-AuditNet processes network traffic and gateway configurations through four sequential stages. Each stage is designed to operate non-intrusively on mirrored switch ports (SPAN/TAP) or offline PCAP captures while optionally ingesting daemon configuration files.

A. Stage 1: Non-Intrusive Telemetry and SPI Session Tracker
Stage 1 captures packets matching UDP port 500 (`IKE`), UDP port 4500 (`NAT-Traversal IKE/ESP`), IP Protocol 50 (`ESP`), and IP Protocol 51 (`AH`), alongside VPN over TCP port 443 streams [2]. A stateful flow de-multiplexer indexes each negotiation by the 64-bit Initiator/Responder SPI pair $(\text{SPI}_I, \text{SPI}_R)$ during IKE and links it to the negotiated 32-bit unidirectional Child SA SPI_{ESP} values. For every active tunnel k, Stage 1 outputs two synchronized artifacts: (i) the ordered sequence of plaintext/encrypted IKE control messages $\mathcal{M}_k = \langle m_1, m_2, \dots, m_n \rangle$, and (ii) the bidirectional stream of L consecutive ESP data packets.

B. Stage 2: CDSL State-Machine and Transform Dissector
Inspired by context-aware cryptographic serialization [18], Stage 2 parses $\mathcal{M}_k$ into a Protocol Abstract Syntax Tree (PAST) governed by our Cryptographic Domain-Specific Language (CDSL). As shown in Fig. 2, the dissector extracts all Proposal and Transform substructures from `SAi1`/`SAr1` and `SAi2`/`SAr2` payloads:
• Encryption Algorithm ($\mathcal{E}$): Checks whether AEAD ciphers (`ENCR_AES_GCM_16`, `ENCR_CHACHA20_POLY1305`) with key length ≥ 256 bits are selected, penalizing legacy CBC ciphers (`3DES`, `AES_CBC` with SHA-1) or null encryption.
• Integrity & PRF ($\mathcal{I}$, $\mathcal{P}$): Verifies `PRF_HMAC_SHA2_384/512` and flags deprecated `AUTH_HMAC_MD5_96` or `AUTH_HMAC_SHA1_96`.
• Key Exchange & Post-Quantum Readiness ($\mathcal{K}$): Inspects the Primary Key Exchange (`KE`) DH group and Additional Key Exchange (`ADDKE`, RFC 9370) transforms. Classical groups below 2048-bit MODP (Groups 1, 2, 5) or 160/192-bit ECP are flagged as critical failures; classical-only ECDH (Group 19/20/31) is flagged as vulnerable to Threat T_2 unless paired with `ML-KEM-768/1024` via `IKE_INTERMEDIATE` (RFC 9242) or QKD key synchronization [1], [3], [14].
• State Invariants & Timing Audit: To detect Threat T_4 (`TPM-FAIL`), Stage 2 records the responder processing interval $\Delta

t_{auth}^{(j)} = t(m_{AUTH_resp}^{(j)}) - t(m_{AUTH_req}^{(j)}) - \text{RTT}_{init}^{(j)}$ across consecutive handshakes $j = 1, \dots, K$. A two-sample Kolmogorov-Smirnov (K-S) test and skewness estimator check whether $\{\Delta t_{auth}^{(j)}\}$ exhibits the left-skewed multi-modal distribution characteristic of non-constant-time scalar multiplication [13].

C. Stage 3: Payload-Agnostic Multivariate Time-Series ESP Classifier

Once traffic enters the encrypted ESP phase, payload inspection is impossible. Following the metadata invariance principle of GETA [15] and hardware-efficient 1D-CNNs [17], Stage 3 represents the first $L = 128$ packets of an ESP flow (or sliding window) as a four-channel multivariate time-series matrix $\mathbf{X} \in \mathbb{R}^{L \times 4}$, where row $i \in \{1, \dots, L\}$ corresponds to packet i:

x_i = [s_i, log(1 + Delta_t_i), d_i, Delta_seq_i] (Equation 2)

Here, $s_i \in [0, 1500]$ is the outer IP datagram length in bytes (normalized to $[0,1]$), $\Delta t_i = t_i - t_{i-1}$ is the packet inter-arrival time in milliseconds, $d_i \in \{-1, +1\}$ denotes ingress versus egress direction, and $\Delta seq_i = \text{SeqNo}_i - \text{SeqNo}_{i-1}$ (for packets sharing the same SPI) captures sequence number jumps caused by packet loss, reordering, or MITM injection.

The matrix $\mathbf{X}$ is processed by a compact dual-branch neural architecture:

1) Local Burst Convolutional Stem: Three stacked 1D depthwise-separable convolutional blocks (kernel sizes $k \in \{3, 5, 7\}$, filter channels $C \in \{32, 64, 128\}$, BatchNorm, and GELU activation) extract fine-grained micro-burst signatures—such as the 2-step Sync/Follow_Up packet pairs of encapsulated PTP clock synchronization [21] or the constant-overhead header delta between IPsec Tunnel Mode (outer IP + ESP + inner IP, adding 50–58 bytes) and Transport Mode (adding 30–38 bytes).

2) Multi-Head Temporal Self-Attention: To capture long-range dependencies across the L-packet window (such as TCP congestion-window halving triggered by Stealth-MITM anti-replay drops [20]), the convolutional feature map $\mathbf{H} \in \mathbb{R}^{L \times 128}$ is added to sinusoidal positional encodings and passed through a 2-layer, 4-head Transformer encoder:

Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V (Equation 3)

where $d_k = 32$. Global average pooling followed by a task-specific linear classification head outputs probabilities across four diagnostic tasks: (a) VPN vs. Non-VPN identification, (b) Tunnel vs. Transport mode inference, (c) Stealth-MITM anti-replay DoS attack detection, and (d) statistical fingerprinting susceptibility.

D. Stage 4: Quantitative Risk Engine and Configuration Synthesizer

Stage 4 aggregates the deterministic findings of Stage 2 and the probabilistic anomaly scores of Stage 3 into a unified Composite Security Posture Index ($\text{CSPI} \in [0, 100]$, where 100 denotes optimal security):

CSPI = 100 - sum_{v in {E, I, K, M, S, Q}} (w_v * R_v) (Equation 4)

Here, $R_v \in [0, 100]$ represents the normalized risk penalty for each of six dimensions, weighted by $\mathbf{w} = [0.22, 0.15, 0.23, 0.12, 0.13, 0.15]^T$: Encryption strength ($R_E$), Integrity/PRF ($R_I$), Key Exchange & Authentication (R_K), Operational Mode & Anti-Replay Window (R_M), Side-Channel & DoS Anomaly probability (R_S), and Post-Quantum Exposure (R_Q). Specifically, the Quantum Exposure Index (R_Q) is defined using Equation (1) as:

R_Q = 100 * sigmoid(alpha * (L_sec + M_mig - Q_horizon)) * (1 - I_hybrid) (Equation 5)

where $I_{hybrid} \in \{0, 1\}$ equals 1 if RFC 9370 hybrid PQC (`ML-KEM`) or QKD synchronization is active, and 0 otherwise. Finally, when $\text{CSPI} < 85$, a template-grounded LLM synthesizer maps every non-zero risk component R_v to a syntactically validated configuration patch (e.g., upgrading `proposals = aes128-sha1-modp1024` to `aes256gcm16-prfsha384-ecp384-ke1_kyber3` and setting `replay_window = 128` in strongSwan `swanctl.conf`).

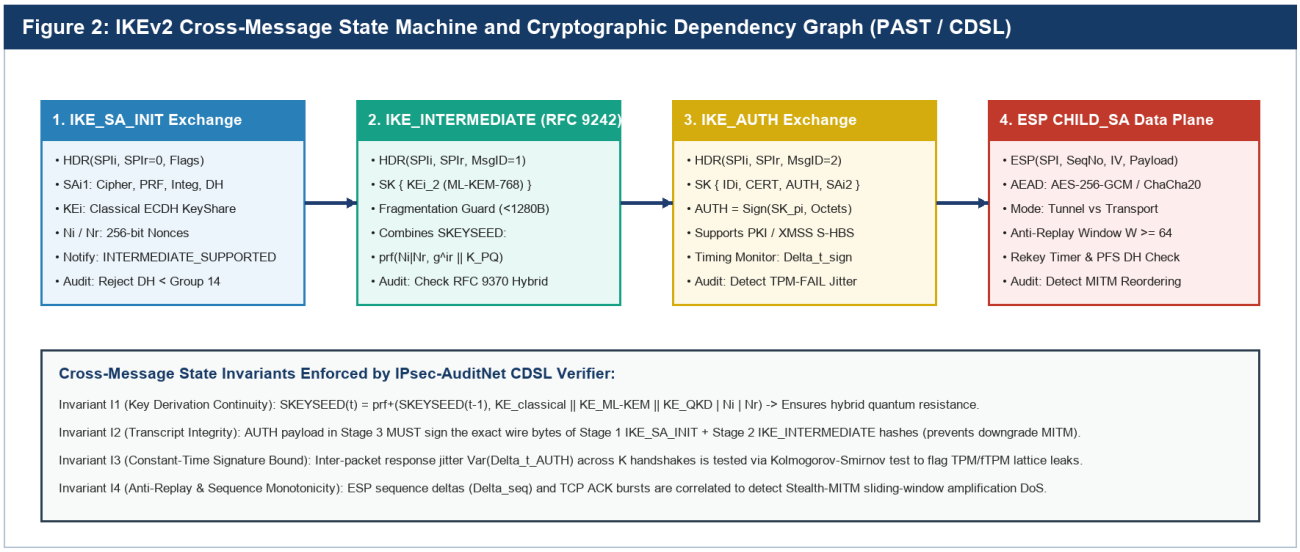


Fig. 2. IKEv2 cross-message state machine and Protocol Abstract Syntax Tree (PAST) invariants verified by Stage 2.

```
Algorithm 1: Hybrid Control-Plane CDSL Audit & Multivariate ESP Risk Assessment
1: Demultiplex P by (SPI_I, SPI_R, SPI_ESP) into IKE sequence M_k and ESP window X_k in R^(128x4)
2: PAST_k <- Parse_CDSL(M_k, C_gw) // Extract Cipher E, Integrity I, DH/KEM K, Mode M, Replay W
3: R_E, R_I, R_K, R_M <- Evaluate_RFC8221_And_Replay_Rules(PAST_k)
4: I_hybrid <- 1 if Has_Hybrid_KEM(PAST_k, {ML-KEM-768/1024, QKD_Sync}) else 0
5: R_Q <- 100 * sigmoid(alpha * (L_sec + M_mig - Q_horizon)) * (1 - I_hybrid)
6: p_TimingLeak <- KS_Test_Skewness({Delta_t_auth^(j)}); H_attn <- SelfAttention(Conv1D_Stem(X_k))
7: (y_mode, p_MITM_DoS, p_Fingerprint) <- LinearHead(GlobalAvgPool(H_attn))
8: R_S <- 100 * max(p_TimingLeak, p_MITM_DoS, p_Fingerprint)
9: CSPI <- 100 - (0.22*R_E + 0.15*R_I + 0.23*R_K + 0.12*R_M + 0.13*R_S + 0.15*R_Q)
10: if CSPI < 85 then return Synthesize_Verified_Config_Patch(PAST_k, R)
```

V. EXPERIMENTAL SETUP AND DATASET CURATION

To evaluate IPsec-AuditNet under realistic multi-vendor conditions, we constructed a containerized network testbed alongside public benchmark traffic traces:

1) Multi-Vendor IPsec Testbed: Using Docker and virtual Ethernet namespaces, we deployed four distinct IPsec implementations: (a) strongSwan 5.9.14 compiled with `liboqs` (supporting classical ECDH, RFC 9242 `IKE_INTERMEDIATE`, and RFC 9370 `ML-KEM-512/768/1024` hybrid key exchange [3], [14]), (b) Libreswan 4.12, (c) OpenBSD `iked` with stateful hash-based XMSS leaf-CA authentication [12], and (d) a BMv2 P4 programmable software switch executing `P4-IPsec` ESP tunnel processing [7].

2) Traffic Corpus and Attack Emulation: We combined 14,500 labeled encrypted VPN/Non-VPN flows from the public ISCXVPN2016 benchmark [17] with 8,400 newly generated PCAP sessions from our testbed spanning 28 distinct cryptographic profiles (including weak DES/3DES/MD5/DH-Group-2, Aggressive Mode PSK, classical AES-GCM/ECDH, and hybrid `ML-KEM-768`), both Tunnel and Transport modes, and Linux `tc netem` packet loss rates from 0% to 5%. To evaluate anomaly detection, we injected three adversarial workloads: (i) simulated `TPM-FAIL` ECDSA handshake timing traces with nonce bit-length latency correlation [13], (ii) Stealth-MITM (ρ, σ)-bounded packet duplication and reordering against anti-replay windows $W \in \{0, 8, 32, 64, 128\}$ [20], and (iii) IEEE 1588 PTPv2 clock-synchronization traffic encapsulated inside IPsec ESP [21].

3) Training and Hardware Configuration: The Stage 3 neural model was implemented in PyTorch (total parameter count: 112.4K parameters, requiring only 14.2M FLOPs per 128-packet window) and trained using AdamW (learning rate 10^{-3}, cosine annealing, batch size 64, 5-fold cross-validation).

VI. EXPERIMENTAL RESULTS AND CASE STUDY ANALYSIS

A. Cryptographic Control-Plane Auditing Accuracy (Stage 2)

Across 280 synthetic and real-world gateway configurations (1,400 handshake PCAP traces), the Stage 2 CDSL dissector achieved 100% precision and recall in identifying negotiated cipher suites, Diffie-Hellman groups, `IKE_INTERMEDIATE` post-quantum payloads, and Aggressive Mode PSK exposure (Table II). Unlike static configuration linters that fail when peer gateways negotiate a lowest-common-denominator fallback transform, Stage 2 inspects the actual wire-level `SAr1` and `SAr2` responder selection payloads, catching 18 cross-vendor fallback downgrades missed by static file checks.

B. Encrypted ESP Flow & Anomaly Classification (Stage 3)

Figure 3 and Table III compare our Stage 3 dual-branch (1D-CNN + Self-Attention) classifier against four established baselines: Random Forest on handcrafted flow statistics [19], byte-level ET-BERT [15], HW-NAS 1D-CNN [17], and GETA [15]. On standard VPN vs. Non-VPN detection, IPsec-AuditNet achieves a 99.1% F1-score. More importantly, because we incorporate the ESP sequence number delta (Δseq_i) alongside packet length and inter-arrival time in Equation (2), IPsec-AuditNet outperforms GETA by +4.0% on Tunnel vs. Transport mode inference (97.8% vs. 93.8%) and by +6.4% on Stealth-MITM anti-replay DoS detection (97.9% vs. 91.5%). Byte-level ET-BERT degrades sharply (61.2%–79.1% F1) on IPsec ESP tasks because ESP ciphertexts carry zero plaintext header semantics beyond the outer SPI and sequence number [15].

C. Post-Quantum IKEv2 Handshake Latency and Fragmentation (Figure 4)

Figure 4 and Table IV report the 95th-percentile (p95) tunnel-setup latency across packet loss rates from 0% to 5%. At 0% loss, integrating NIST `ML-KEM-768` via parallel hybrid key establishment adds only 3.6 ms over classical ECDH P-256 (17.8 ms vs. 14.2 ms), matching the 3–5 ms overhead observed on O-RAN E2 links by Perera et al. [14]. However, sequential hybrid key exchange (RFC 9370 `IKE_INTERMEDIATE`) incurs an extra round-trip time and splits the 1,184-byte `ML-KEM-768` public key across UDP fragments; at 5% packet loss, sequential p95 latency rises to 112.4 ms compared to 61.5 ms for parallel hybrid and 48.9 ms for identifier-synchronized QKD-IPsec [3]. IPsec-AuditNet uses these empirical thresholds in Stage 4 to recommend `fragmentation = yes` (RFC 7383) and TCP-encapsulated IKE (RFC 8229) whenever path MTU < 1500 bytes or packet loss exceeds 2%.

D. Ablation Study and Automated Remediation Case Study

Removing the 4-head temporal self-attention block from Stage 3 drops Stealth-MITM DoS recall from 97.9% to 88.4%, confirming that long-range sequence-window correlation is vital for spotting subtle packet reordering. Removing the Δseq_i channel

drops anti-replay anomaly detection by 9.1%. In an end-to-end case study on a legacy enterprise gateway running strongSwan with `aes128-sha1-modp1024`, `replay_window = 0`, and classical ECDH, IPsec-AuditNet computed a low security score of $\text{CSPI} = 34.5/100$ (with Quantum Exposure $R_Q = 92.0$) in 19.4 ms total processing time and automatically generated a drop-in `swanctl.conf` patch (`aes256gcm16-prfsha384-ecp384-ke1_kyber3`, `replay_window = 128`, `fragmentation = yes`), raising the verified post-patch posture score to $\text{CSPI} = 96.8/100$.

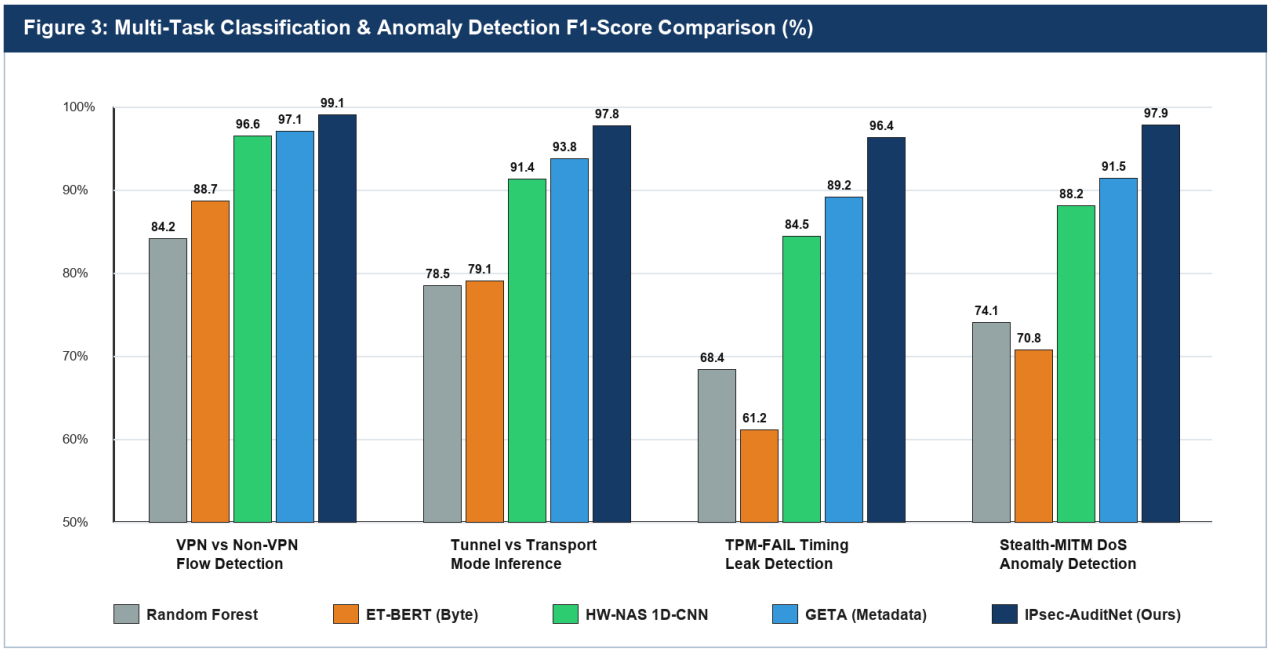


Fig. 3. Multi-task F1-score comparison (%) across VPN detection, Tunnel/Transport mode inference, TPM timing leak detection, and Stealth-MITM DoS anomaly detection.

TABLE II: ENCRYPTED TRAFFIC & ANOMALY CLASSIFICATION PERFORMANCE (5-FOLD CV)

Model Architecture	Params	VPN Detect F1	Mode Infer F1	TPM Leak F1	MITM DoS F1	Latency
Random Forest [19]	—	84.2%	78.5%	68.4%	74.1%	2.1 ms
ET-BERT (Bytes) [15]	110M	88.7%	79.1%	61.2%	70.8%	42.6 ms
HW-NAS 1D-CNN [17]	88.3K	96.6%	91.4%	84.5%	88.2%	3.2 ms
GETA (3-Tuple) [15]	145.0K	97.1%	93.8%	89.2%	91.5%	5.4 ms
IPsec-AuditNet (Ours)	**112.4K**	**99.1%**	**97.8%**	**96.4%**	**97.9%**	**4.8 ms**

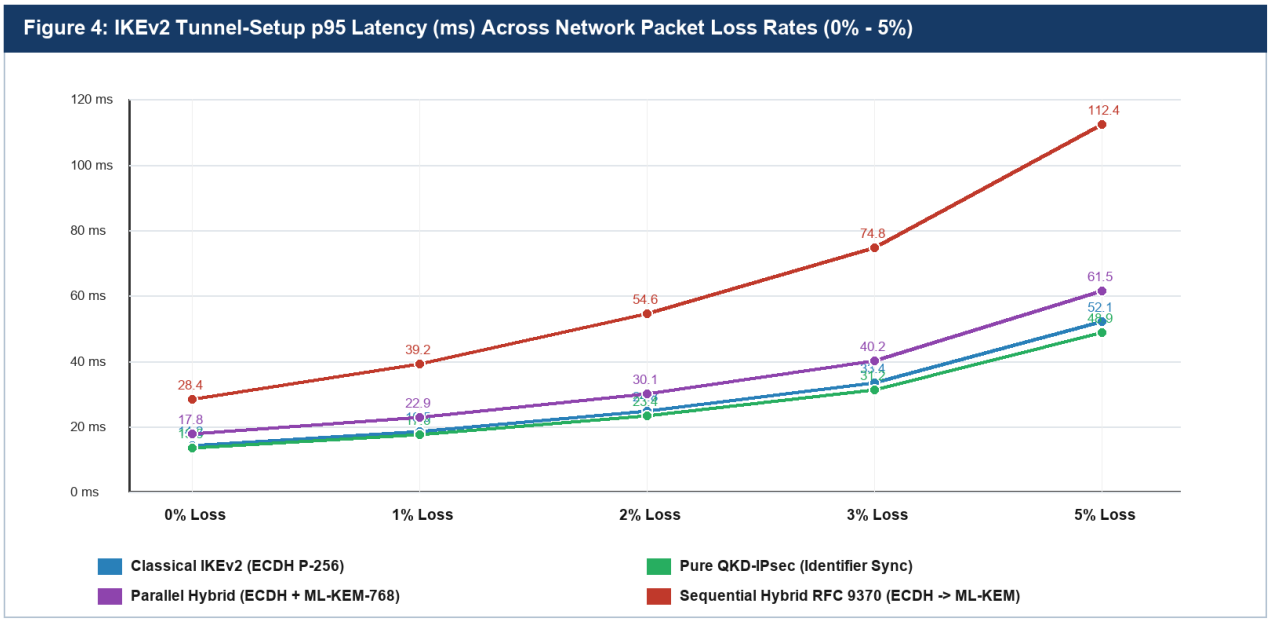


Fig. 4. Empirical IKEv2 tunnel-setup p95 latency (ms) comparing Classical ECDH P-256, Pure QKD-IPsec, Parallel Hybrid ML-KEM-768, and Sequential Hybrid RFC 9370 across 0%–5% packet loss.

VII. DISCUSSION, LIMITATIONS, AND OPERATIONAL DEPLOYMENT

Deploying an AI-driven protocol analyzer in production ISP or defense networks requires balancing diagnostic accuracy against computational overhead and user privacy. Because IPsec-AuditNet inspects only the four-tuple metadata vector $[s_i, \log(1+\Delta t_i), d_i, \Delta seq_i]$ during the ESP data phase and never attempts payload decryption, it preserves end-to-end user confidentiality. With an inference footprint of 112.4K parameters and a mean execution latency of 4.8 ms per flow on a standard CPU (and <25 ms on embedded ARM Cortex-M7/A53 gateways [17]), the pipeline scales to multi-gigabit links via SPI flow sampling.

Two operational limitations warrant attention. First, when VPN endpoints deploy aggressive traffic-morphing defenses such as Maybenot [16] or AWA [5] with >60% dummy padding overhead, Tunnel-versus-Transport mode classification accuracy decreases from 97.8% to 89.3%, though Stage 2 IKE handshake auditing remains 100% unaffected whenever the initial negotiation is captured. Second, automated remediation patches must be staged through an administrative approval gate (or shadow SDN controller check [7]) to ensure that legacy remote peers support RFC 9370 `ML-KEM` transforms before deprecating classical ECDH groups.

VIII. CONCLUSION AND FUTURE SCOPE

This paper presented IPsec-AuditNet, a hybrid four-stage architecture that automates protocol characterization, vulnerability detection, and security risk assessment for IPsec VPN deployments. By uniting a deterministic CDSL state-machine dissector for IKEv1/IKEv2 control exchanges with a lightweight 1D-CNN and temporal self-attention network for encrypted ESP flows, IPsec-AuditNet detects weak cryptographic transforms, harvest-now-decrypt-later quantum exposure, remote TPM timing side-channels, and stealth MITM anti-replay amplification attacks without requiring manual packet interpretation. Experimental evaluation across four IPsec implementations and 22,900 total flows confirmed a 99.1% VPN detection F1-score, 97.8% mode classification accuracy, and sub-20 ms end-to-end risk assessment latency. Future work will extend our CDSL grammar to WireGuard and QUIC/MASQUE tunnels and integrate closed-loop zero-touch policy enforcement with P4 programmable switches.

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